Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/14/2006
PI Name:	Garoff, Stephen Ph.D.		
Project Title:	Wetting Dynamics of Complex Fluids and Surfaces		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sg2e+@andrew.cmu.edu	**Fax:**	FY 412-681-0648
PI Organization Type:	UNIVERSITY	**Phone:**	412-268-6877
Organization Name:	Carnegie Mellon University		
PI Address 1:	Department of Physics		
PI Address 2:			
PI Web Page:			
City:	Pittsburgh	**State:**	PA
Zip Code:	15213-3890	**Congressional District:**	14
Comments:	T		
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	12/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	4	**No. of Master' Degrees:**	4
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA GRC
Contact Monitor:	Jacqmin, David	**Contact Phone:**	216-433-5853
Contact Email:	fsdavid@tess.grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Walker, Lynn (Carnegie Mellon University) Rame, Enrique (National Center for Microgravity Research)		
Grant/Contract No.:	NNC04GA34G		
Performance Goal No.:	2.7		
Performance Goal Text:			
Task Description:	Using very accurate optical imaging techniques and particle image velocimetry, we characterize the shape of the fluid/fluid interface and the flow fields in a microscopic region near moving contact lines. From these images, we test models of the hydrodyna mics in regions near the contact line where viscous forces dominate the fluid flow and in regions even closer to the contact line where a unique hydrodynamic regime must govern the fluid motion. By making systematic changes in the materials system under examination, we seek the physical origins of these unique hydrodynamics. Low gravity studies are essential to this program; bouyancy flows that would otherwise mask the effects of capillary and wetting forces are eliminated. The study will characterize the unique hydrodynamics occurring in a microscopic region near moving contact lines and that controls macroscopic spreading of fluids on solid surfaces.		

	The task significance is to (1) establish the correct method for measuring dynamic contact angle boundary condition, (2) provide a proper basis for predictive models of spreading, (3) test models of the hydrodynamics near, but not at, moving contact lines where viscous forces are dominant, and (4) identify the molecular-scale mechanisms governing the unique hydrodynamic regimes at moving contact lines.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research not only elucidates the fundamental hydrodynamics which control wetting by surfactant solutions adn elastic fluids; it also provides the understanding needed to optimize and control the many technological processes, such as cleaning and coating or printing, which employ dynamic wetting by surfactant solutions and non-Newtonian fluds.
Task Progress:	During the past year, we have investigated dynamic wetting by two classes of complex fluids: surfactant solutions and polymer melts. Our research not only elucidates the fundamental hydrodynamics which control wetting by surfactant solutions; it also provides the understanding needed to optimize and control the many technological processes, such as cleaning and coating, which employ dynamic wetting by surfactant solutions. We have experimentally investigated the hydrodynamics of receding contact lines and surfactant deposition from soluble surfactant solutions in two velocity regimes: at low substrate withdrawal speeds where the surface emerges dry, and at higher withdrawal speeds where a film is entrained on the surface. We have developed a technique to measure the fluid velocity on the liquid-vapor interface and this enables us to quantitatively study the Marangoni stresses and surfactant transport near the contact line. We determine the Marangoni stresses near the contact line. While they are the least dominant stresses, they do alter the flow fields compared to those of a pure fluid. The surfactant-surface interactions at the contact line govern the Marangoni stresses and thereby control the hydrodynamics. In the two velocity regimes examined, we observe two different mechanisms of surfactant deposition on the solid-vapor interface. Island structures with quantized heights are found and the structure of these islands can be varied by changing the withdrawal and evaporation rates. We also see a sharp transition in the boundary between these two wetting regimes with surfactant concentration. We find that this transition is universal across several surfactant-surface systems. Our studies on a series of non-Newtonian fluids suggest looking at dynamic wetting from a new perspective. Flow in the wedge-like region near the moving contact line demands that shear rates increase to very high values as the contact line is approached. Thus, somewhere in the flow field, molecular relaxations in the fluid will be unable to relax, leading to non-Newtonian behavior on some small scale near the contact line. This is crucial because in dynamic wetting, behavior at a very small, micron scale has impact on wetting behavior at the submillimeter and millimeter scale where fluid behavior must be precisely controlled in many applications. We find that non-Newtonian fluids dominated by shear thinning reduce viscous bending near the contact line while fluids dominated by elasticity increase the bending, compared to a Newtonian fluid with the same zero shear viscosity. These changes in the viscous bending can, in turn, impact the dependence of the dynamic contact angle on capillary number (Ca), changing it from the classic Ca1/3 exhibited by Newtonian fluids. We find that even fluids with such short relaxation times and undetectable non-Newtonian behavior in their rheological characterization, do not show dynamic wetting described by models that assume Newtonian behavior. Our work suggests a modified flow field with a region of non-Newtonian behavior intervening between the region at shortest length scales where the classical contact line singularity is alleviated and the region far from the contact line where viscous forces just begin to become significant. Technological Impact: These studies will develop a complete picture of the hydrodynamics that control wetting of non-Newtonian fluids. They will establish the principles for design of the many technological processes involving wetting by non-Newtonian fluids.
Bibliography Type:	Description: (Last Updated: 07/02/2007)
Abstracts for Journals and Proceedings	Garoff S, Rame E. "Dynamic wetting by soluble surfactant solutions." Presented at 13th International Coating Science and Technology Symposium, Denver, CO, September 2006. Abstract book, 13th International Coating Science and Technology Symposium, September 2006. , Sep-2006
Abstracts for Journals and Proceedings	Ramé E, Walker L, Garoff S. "Spreading of liquids on solid surfaces: Pure fluids." Presented at 13th International Coating Science and Technology Symposium, Denver, CO, September 2006. Abstract book, 13th International Coating Science and Technology Symposium, September 2006. , Sep-2006
Abstracts for Journals and Proceedings	Seevaratnam G, Garoff S, Rame E, Walker LM. "Excitation of non-Newtonian behavior in fluids in the wedge flow near moving contact line." American Chemical Society (ACS), Colloid and Surface Science Symposium, Boulder, CO, June 2006. Abstract book, American Chemical Society (ACS), Colloid and Surface Science Symposium, June 2006. , Jun-2006
Abstracts for Journals and Proceedings	Varanasi K, Beppler B, Rame E, Garoff S. "Self-assembled surfactant structures emerging from films and contact lines of evaporating solvents." Presented at ACS, Colloid and Surface Science Symposium, Boulder, CO, June 2006. Abstract book, American Chemical Society (ACS), Colloid and Surface Science Symposium, June 2006. , Jun-2006
Abstracts for Journals and Proceedings	Seevaratnam G, Garoff S, Rame E, Walker LM. "The impact of fluid behavior on the wedge-flow near moving contact lines." Presented at AIChE (American Institute of Chemical Engineers) Annual Meeting, San Francisco, CA, November 2006. Online abstracts, AIChE (American Institute of Chemical Engineers) Annual Meeting, November 2006. , Nov-2006
Articles in Peer-reviewed Journals	Chen X, Rame E, Garoff S. "The effects of thin and ultrathin liquid films on dynamic wetting." Physics of Fluids, 2004 Feb;16(2):287-97. http://dx.doi.org/10.1063/1.1632498 , Feb-2004
Articles in Peer-reviewed Journals	Rame E, Garoff S, Willson KR. " Characterizing the microscopic physics near moving contact lines using dynamic contact angle data." Phys Rev E Stat Nonlin Soft Matter Phys. 2004 Sep;70(3 Pt 1):031608. Epub 2004 Sep 29. PMID: 15524535 , Sep-2004

Articles in Peer-reviewed Journals	Seevaratnam GK, Walker LM, Rame E, Garoff S. " Wetting by simple room-temperature polymer melts: deviations from Newtonian behavior." J Colloid Interface Sci. 2005 Apr 1;284(1):265-70.PMID: 15752812 , Apr-2005
Presentation	E. Rame "The Influence of Dynamic Contact Angle on wetting Dynamic" ALA Lab Automation Jan-2004
Presentation	G. Seevaratnam, L.M. Walker, S. Garoff and E. Rame "Wetting by Simple Room Temperature Polymer Melts: Deviations from Newtonian Behavior" 78th ACS Colloids and Surface Science Symposium Jun-2004
Presentation	K S. Varanasi, S. Garoff "Concentration Dependent Transition in the Critical Velocity of Film Pulling in Surfactant Solutions" 78th ACS, Colloid and Surface Science Symposium, New Haven Jun-2004
Presentation	K S. Varanasi, S. Garoff "Concentration Dependent Transition in the Critical Velocity of Film Pulling in Surfactant Solutions" 78th ACS, Colloid and Surface Science Symposium, New Haven Jun-2004
Presentation	E.Rame "Modelado de Mojado de Sólidos por Líquidos: un enfoque Fluido-mecánico" Grupo de Medios Porosos (Porous Media Group), Physics Dept. of the School of Engineering, University of Buenos Aires, Buenos Aires Aug-2004
Presentation	E.Rame "Modeling Liquid Spreading: A Fluid Mechanical Approach" Dept. Mechanical Engineering, Michigan Technological Univ Sep-2004
Presentation	K, Varanasi, S. Conroy, S. Garoff, B. Luokkala, E. Rame "Dynamic Wetting by Soluble Surfactant Solutions" Interantional Conference on Wetting, Hiafa Israel Jul-2005
Presentation	E. Rame, G. Seevaratnem, L. Walker, S. Garoff "Dynamic Wetting of Room Temperature Polymer Melts" NASA Fluids Meeting Jun-2004
Presentation	Gitanjali K. Seevaratnam, Steve Garoff, Lynn M. Walker, Enrique Rame "Influence of elasticity on the dynamic interface shapes of polymer melts and Boger fluids" Soc. Of Rheology, 76th Annual Meeting Feb-2005
Presentation	Gita Seevaratnam, Stephen Garoff , Lynn M. Walker, Enrique Rame "Influence of Non-Newtonian Behavior on the Dynamic Interface Shapes of Polymer Melts, Solutions and Boger Fluids" 79th ACS Colloid and Surface Science Symposium Jun-2005
Presentation	E. Rame, K. Varanasi, B. Luokkala, S. Conroy, S. Garoff "The spreading of surfactant solutions. Effect of surfactant mass transfer condition at the contact line" Interdisciplinary Transport phenomena in microgravity and space sciences“, Tomar, Portugal, August 7-12 2005 Aug-2005
Presentation	E. Rame, K. Varanasi, B. Luokkala, S. Conroy, S. Garoff "Wetting by surfactant solutions: The role of mass transfer through the contact line" APS-DFD, Chicago Nov-2005
Presentation	E. Rame, K. Varanasi, B. Luokkala, S. Conroy, S. Garoff "Dynamic Wetting by Soluble Surfactant Solutions" Rutgers University Sep-2005
Presentation	S. Garoff, E. Rame, K. Varanasi, B. Luokkala, S. Conroy "Dynamic Wetting by Soluble Surfactant Solutions" Kent State University Jan-2005